

A translational tile with no fully periodic tiling in dimension three

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Abstract

We construct a finite translational tile in $\mathbb{Z}^3$ that admits tilings but no fully periodic tiling. Its unit-cube thickening has the same property in $\mathbb{R}^3$, even when arbitrary real translations are allowed. This gives a negative resolution of the periodic tiling conjecture in dimension three.

1 Introduction

For subsets A, F of an abelian group G , write $A \oplus F = G$ when every element of G has a unique expression $a + f$ with $a \in A$ and $f \in F$. A finite nonempty set F is a *translational tile* if such a set A exists. A tiling is *fully periodic* if A is invariant under a subgroup of finite index in G . In $\mathbb{R}^3$, a bounded measurable set of positive measure tiles by translations when its translates partition space up to null sets; full periodicity means that its translation set is invariant under a lattice of full rank. Throughout this paper, “periodic” means fully periodic.

The periodic tiling conjecture asks whether every translational tile admits a periodic tiling. An explicit Euclidean formulation appears in Lagarias and Wang [7, p. 346]. Our conclusion in dimension three is the following.

Theorem 1.1. *There is a finite nonempty set $T \subset \mathbb{Z}^3$ such that:*

- (i) *T tiles $\mathbb{Z}^3$ by translations, but no translation set A with $A \oplus T = \mathbb{Z}^3$ is invariant under a subgroup of finite index in $\mathbb{Z}^3$;*
- (ii) *the set $\Omega = T + [0, 1]^3$ tiles $\mathbb{R}^3$ by translations up to null sets, but no such tiling has a translation set invariant under a lattice of full rank in $\mathbb{R}^3$.*

The distinction between a tile and a particular tiling is essential: a tile may admit both periodic and nonperiodic translation sets. The theorem excludes every periodic translation set for the constructed tile. For lattice tilings, dimension three is the smallest possible dimension for this conclusion. Newman’s finite-state argument shows that the translation set of every tiling of $\mathbb{Z}$ by a finite tile is periodic [9, proof of Theorem 2]. Bhattacharya proved that every finite tile of $\mathbb{Z}^2$ admits a periodic tiling [1, Theorem 1.1]. Greenfeld and Tao subsequently gave a quantitative version of the planar result and proved that the translation set of every tiling of $\mathbb{Z}^2$ by one finite tile is a disjoint union of finitely many sets, each invariant under some nonzero translation [3, Theorems 1.3(i) and 1.5]. An earlier planar result of Wijshoff and van Leeuwen [10, Theorem 5.4] concerns polyominoes without holes and produces a tiling whose translation set is a single lattice. The unrestricted finite-tile theorem allows arbitrary disconnected sets and only asks for a finite-index period subgroup. For Euclidean tilings, de Dios Pont et al. [2, Theorem 1.1] prove periodic-tiling existence for rational polygonal sets, including disconnected sets and sets with holes. These positive

planar results do not establish the conjecture for arbitrary bounded measurable Euclidean tiles. We use the lattice results only for the least-dimension conclusion above; the Euclidean assertion of [Theorem 1.1](#) is proved directly below.

Greenfeld and Tao [\[5\]](#) disproved the periodic tiling conjecture in sufficiently large dimension. Their construction first produces a tile in $\mathbb{Z}^2 \times G_0$, with G_0 a finite abelian 2-group, by encoding a Sudoku rule with an arithmetic obstruction to periodicity. The present proof follows this general strategy and the p -adic Sudoku formulation of Greenfeld and Tao [\[6, Section 4\]](#). We also use the idea of combining simultaneous tiling equations into one equation from Greenfeld and Tao [\[5, Section 3\]](#). The quotient-to-lattice reduction of Meyerovitch et al. [\[8, Theorem 1.1 and Corollary 1.2\]](#) explains the relevance of controlling the finite factor: $\mathbb{Z}^2$ times a finite cyclic group is a quotient of $\mathbb{Z}^3$.

Here the Sudoku object is an array $W(n, m)$ over a finite alphabet, with n in a finite interval I of columns and $m \in \mathbb{Z}$ indexing rows. A finite set of allowed words prescribes the possible samples $(W(n, dn + e))_{n \in I}$ along every integer-slope line. The arithmetic word rule has two features: one such array exists with every column nonconstant, but no array with those properties has a vertical period. We turn these finite word constraints and column conditions into simultaneous tiling equations.

There are two requirements at each stage of this strategy. The encoding must admit at least one tiling, and every fully periodic tiling of the encoded object must yield a fully periodic solution of the preceding constraints. A system of tiling equations therefore means several equations with the *same* unknown translation set. Separate solutions of the equations would not suffice. The finite group matters as well: a general finite abelian group needs several generators, whereas a cyclic group can be represented by just the third integer coordinate.

The central construction here encodes the Sudoku constraints while keeping that finite factor cyclic. For each word position, a finite-valued function has one input digit designated for each symbol. A symbol is called active when changing that digit can change the function's value. An arbitrary solution may activate several symbols at one position; the constraints are imposed on every resulting choice of symbols. The explicit solution we construct activates just one. Cycle tests exclude forbidden simultaneous dependencies, and activation tests force at least one symbol at every ordinary position. Two seed channels force distinct values in every column of the resulting Sudoku array. The activation tests use a nonuniform list of output shifts with uniform coordinate counts on a product of two prime-order groups. These counts force activity, while suitable permutations of the remaining coordinates give an explicit common solution to all the equations. Both seed tests use the same auxiliary output function. All finite group factors have pairwise coprime orders, so their product remains cyclic. The horizontal residue pairs needed by the seed tests only index values of one cyclic output coordinate; they do not become an additional group factor.

A stacking construction using a fresh prime then gives one tile in $\mathbb{Z}^2 \times \mathbb{Z}/Q\mathbb{Z}$. We supply a direct rigidification argument that passes to $\mathbb{Z}^3$ and also handles the unit-cube thickening in $\mathbb{R}^3$. Integer vertices alone would not justify restricting an arbitrary Euclidean tiling to integer translations. Our construction uses two component shapes with the same large common part and one displaced marked cube. The common part forces the component centers in a hypothetical periodic tiling onto a translated grid. Coverage of the marked cubes then forces invariance in the quotient kernel, allowing that tiling to descend. Thus the Euclidean argument includes arbitrary real translation vectors. The result concerns full periodicity; neither connectedness of the tile nor absence of every individual nonzero period is asserted.

[Figure 1](#) records the two directions maintained by these constructions. The proof is organized as follows. [Section 2](#) proves the Sudoku obstruction and gives its last-nonzero-digit model. [Section 3](#) constructs the cyclic group and the dependence tests; [Section 4](#) forces activity and rules out periodic common solutions. [Section 5](#) constructs one common solution explicitly. [Sections 6](#) and [7](#) perform stacking and the passage to three dimensions. Every construction and nonperiodicity argument

needed for [Theorem 1.1](#) is proved in the paper.

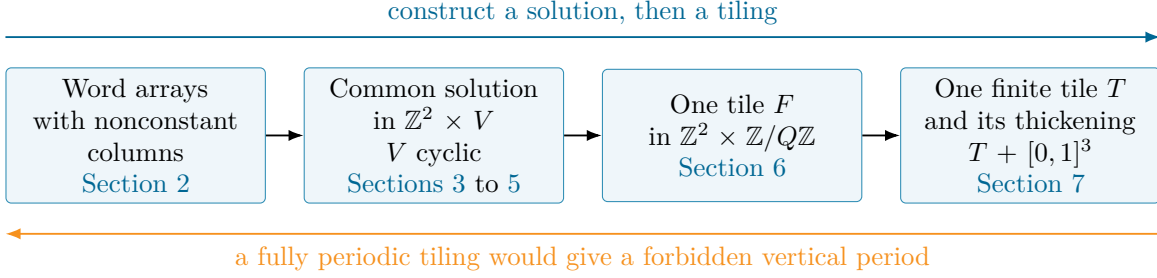


Figure 1: The two directions of the proof. The construction proceeds to the right, starting from the explicit last-nonzero-digit array in [Theorem 2.5](#). To exclude full periodicity, any hypothetical periodic tiling is decoded in the reverse direction, producing a line-rule array with nonconstant columns and a vertical period. Every tiling equation at the second stage uses the same translation set, and the last stage uses the same T in both the discrete and Euclidean assertions.

2 A word rule with no vertical period

We begin with a finite word constraint whose solutions on an infinite strip cannot have a vertical period if every column is nonconstant. The constraint is the two-digit p -adic Sudoku rule of Greenfeld and Tao [6, Definition 4.5]. The affine-square and rescaling arguments below follow the framework of Greenfeld and Tao [6, Theorem 4.7, Lemma 4.8, and Section 4.2]; we give the complete argument for the particular obstruction needed here.

Fix a prime $p > 200$, and set

$$N = p^2, \quad \Sigma = \mathbb{F}_p^\times, \quad I = \{0, 1, \dots, N - 1\}.$$

When an integer occurs in an expression over $\mathbb{F}_p$, its residue modulo p is understood.

Definition 2.1 (Allowed words and the line rule). A word $w: I \rightarrow \Sigma$ is *allowed* if there are integers a, b , not both divisible by p , such that, for every $n \in I$,

$$w(n) = \begin{cases} an + b \pmod{p}, & p \nmid an + b, \\ (an + b)/p \pmod{p}, & p \mid an + b \text{ and } p^2 \nmid an + b. \end{cases}$$

At indices with $p^2 \mid an + b$, the value may be any member of Σ . An array $W: I \times \mathbb{Z} \rightarrow \Sigma$ satisfies the *line rule* if

$$n \mapsto W(n, dn + e)$$

is an allowed word for every $d, e \in \mathbb{Z}$. A *positive vertical period* of W is an integer $M > 0$ such that $W(n, m + M) = W(n, m)$ for all $(n, m) \in I \times \mathbb{Z}$.

The allowed-word constraint depends only on the residues of a, b modulo p^2 . Indeed, these residues determine the divisibility alternatives and each of the prescribed values. In particular, it is a finite constraint on words over the finite alphabet Σ .

For every allowed word, reduction of a witnessing pair gives a nonzero affine form $\ell(n) = \bar{a}n + \bar{b}$ over $\mathbb{F}_p$. The word agrees with ℓ wherever $\ell \neq 0$. A nonzero affine form has at most one zero residue class, so there are at most $N/p = p$ exceptional indices on I . Here and below, “nonzero affine form” means that its coefficients are not all zero; a nonzero constant form is permitted.

Lemma 2.2 (Global affine approximation). *If W satisfies the line rule, there are $A, B, C \in \mathbb{F}_p$, not all zero, such that*

$$W(n, m) = An + Bm + C \quad \text{whenever } An + Bm + C \neq 0.$$

Proof. For each line $m = dn + e$ with $d \in \{-1, 0, 1\}$, choose the nonzero affine form $\ell_{d,e}(n)$ supplied by an allowed-word witness. Call an index on that line exceptional when $\ell_{d,e}(n) = 0$.

First we find a consecutive 4×4 square containing no exceptional cell for any of these three line directions. In the rectangle $I \times \{0, \dots, K-1\}$ there are $(N-3)(K-3)$ such squares. The numbers of lines of slopes $0, 1, -1$ meeting the rectangle are respectively $K, K+N-1, K+N-1$. Each line has at most p exceptional cells on the whole strip, and each cell belongs to at most 16 candidate squares. Consequently at most

$$16p(3K + 2N - 2) \leq 16p(3K + 2N)$$

squares are excluded. Since

$$N - 3 - 48p = p^2 - 48p - 3 > 0,$$

we can choose an integer $K \geq 4$ with

$$K > \frac{3(N-3) + 32pN}{N-3-48p}.$$

For this choice,

$$(N-3)(K-3) - 16p(3K+2N) = K(N-3-48p) - 3(N-3) - 32pN > 0.$$

A square of the required kind therefore exists.

Write its column indices as $r, r+1, r+2, r+3$ and its row indices as $q, q+1, q+2, q+3$. Each row agrees on the square with an affine function $U_m n + V_m$. For $m = q+1, q+2$ and $n = r+1, r+2$, the second difference along either diagonal vanishes. Thus, for each sign,

$$0 = n(U_{m+1} - 2U_m + U_{m-1}) + (V_{m+1} - 2V_m + V_{m-1}) \\ \pm (U_{m+1} - U_{m-1}).$$

Subtracting the identities at the two consecutive values of n gives $U_{m+1} - 2U_m + U_{m-1} = 0$. Subtracting the two signs gives $U_{m+1} = U_{m-1}$, because $2 \neq 0$ in $\mathbb{F}_p$. Together these imply $U_{m-1} = U_m = U_{m+1}$. The two overlapping triples show that all four row slopes have a common value A . The remaining identities say that the four intercepts have zero second differences, so they have the form $V_m = Bm + C$. Therefore the form

$$H(n, m) = An + Bm + C$$

agrees with W on the square. It is not the zero form, because every entry of W lies in Σ .

Call a cell *good* if either $H(n, m) = 0$ or $W(n, m) = H(n, m)$. We claim that four consecutive good cells on a line of slope $0, 1$, or -1 make the entire line good. The restriction of H to that line is an affine form in n . If it is identically zero, the claim follows from the definition of goodness. Otherwise, it and the chosen form $\ell_{d,e}$ each have at most one zero among the four indices, which are distinct modulo p . At least two of the four indices remain where both forms are nonzero. At these indices, goodness and the word rule identify both forms with W . Two distinct residues determine an affine form, so the two forms are equal. Wherever this common form is nonzero the word rule gives $W = H$, and its zeros are good by definition. This proves the claim.

Applying the claim to the four square rows makes those entire rows good. To adjoin a row immediately above a block of four good rows, consider its cell in column n . At least one of

$$n + 4 \leq N - 1 \quad \text{or} \quad n - 4 \geq 0$$

holds, since $N \geq 8$. Choose $\varepsilon \in \{-1, 1\}$ so that the four columns $n + \varepsilon, n + 2\varepsilon, n + 3\varepsilon, n + 4\varepsilon$ are in I . If the new row has index m , the cells

$$(n + j\varepsilon, m - j), \quad 1 \leq j \leq 4,$$

are four consecutive good cells on a diagonal through (n, m) . The claim makes (n, m) good. For a row immediately below a block of four good rows, use $(n + j\varepsilon, m + j)$ instead. These formulas include both edge columns of the strip. Successively adjoining rows in both directions makes every cell good, proving the lemma. $\square$

Lemma 2.3 (Vertical nondegeneracy). *Let $H(n, m) = An + Bm + C$ be any affine form supplied by Theorem 2.2. Every column of W is nonconstant if and only if $B \neq 0$. In that case every positive vertical period of W is divisible by p .*

Proof. If $B = 0$, then $An + C$ is not identically zero. Some $n \in \{0, \dots, p - 1\}$ therefore has $An + C \neq 0$, and the entire column at that n has the constant value $An + C$. Conversely, if $B \neq 0$, then for any fixed n the form $An + Bm + C$ runs through $\mathbb{F}_p$ as m runs through a complete residue system modulo p . At its $p - 1$ nonzero values, the entries of W equal those values. Thus every column takes every symbol of Σ , and in particular is nonconstant.

Suppose now that $M > 0$ satisfies $W(n, m + M) = W(n, m)$ for every (n, m) . For a fixed n , choose m so that both $H(n, m)$ and $H(n, m + M)$ are nonzero. There are at most two excluded residue classes, so this is possible. Equality of the two entries gives

$$BM = H(n, m + M) - H(n, m) = 0 \quad \text{in } \mathbb{F}_p.$$

Since $B \neq 0$, this implies $p \mid M$. $\square$

Proposition 2.4 (The vertical-period obstruction). *There is no array $W : I \times \mathbb{Z} \rightarrow \Sigma$ satisfying the line rule, having every column nonconstant, and having a positive vertical period.*

Proof. Suppose otherwise, and choose the smallest positive integer M that occurs as a vertical period of any such array. Choose an array with period M and an affine approximation $An + Bm + C$. By Theorem 2.3, $B \neq 0$ and $p \mid M$.

Choose integers u, v with $A + Bu = C + Bv = 0$ in $\mathbb{F}_p$, and replace W by

$$\widetilde{W}(n, m) = W(n, m + un + v).$$

On a line of slope d and intercept e , this samples the original line of slope $d + u$ and intercept $e + v$, so the line rule is preserved. Each column is simply translated in its row coordinate; hence column nonconstancy and the set of vertical periods are preserved. The new affine approximation is Bm . Rename this normalized array W .

Define

$$W_1(n, m) = W(n, pm).$$

This array also satisfies the line rule: its line with slope d and intercept e is the old line with slope pd and intercept pe . By Theorem 2.2, it has a nonzero affine approximation $H_1(n, m) = A_1n + B_1m + C_1$. We next prove $B_1 \neq 0$, which is the step needed to retain column nonconstancy under this rescaling.

If $B_1 = 0$, choose $0 \leq n_0 < p$ with $c = A_1 n_0 + C_1 \neq 0$. Apply the word rule for W to the line $m = n - n_0$, and let a, b be a witnessing pair. Away from the at most two zero residue classes of $\bar{a}n + \bar{b}$ and $B(n - n_0)$, both forms equal the array entry on that line. Since $p - 2 \geq 2$, they agree at two distinct residues and therefore are the same affine form. In particular,

$$a \equiv B \pmod{p}, \quad b \equiv -Bn_0 \pmod{p}.$$

It follows that $p \mid an_0 + b$ and $p \nmid a$.

For $0 \leq k < p$, put $n = n_0 + pk$; all these indices lie in I , including the largest possible value $p^2 - 1$. At the corresponding point of the line, the entry is

$$W(n_0 + pk, pk) = W_1(n_0 + pk, k) = c,$$

where the last equality follows from the nonzero value $H_1(n_0 + pk, k) = c$. On the other hand,

$$a(n_0 + pk) + b = p \left(\frac{an_0 + b}{p} + ak \right).$$

Whenever the parenthesized expression is nonzero modulo p , the second case of the allowed-word rule forces this entry to equal

$$\frac{an_0 + b}{p} + ak \pmod{p}.$$

As k runs from 0 to $p - 1$, these residues run through all of $\mathbb{F}_p$, because $p \nmid a$. Thus $p - 1$ of the entries are forced to take the $p - 1$ distinct nonzero values, contradicting their common value c . The remaining index, where the rule prescribes no value, is not needed for this contradiction.

We have proved $B_1 \neq 0$, so every column of W_1 is nonconstant by [Theorem 2.3](#). But M/p is a positive vertical period of W_1 , since

$$W_1(n, m + M/p) = W(n, pm + M) = W(n, pm) = W_1(n, m).$$

This contradicts the choice of M . □

Lemma 2.5 (A model for the line rule). *For $t \in \mathbb{Z} \setminus \{0\}$, let $\text{ord}_p(t)$ be the largest nonnegative integer r for which $p^r \mid t$, and define*

$$f(t) = \frac{t}{p^{\text{ord}_p(t)}} \pmod{p}, \quad f(0) = 1.$$

Then $f: \mathbb{Z} \rightarrow \Sigma$, and the array $W(n, m) = f(m)$ satisfies the line rule and has every column nonconstant.

Proof. The definition gives a nonzero residue for every nonzero integer, including negative integers. For every $r \geq 0$ and $t \in \mathbb{Z}$ we have $f(p^r t) = f(t)$; this also holds at $t = 0$ by definition.

Fix integers d, e . If $(d, e) \neq (0, 0)$, let r be the largest integer such that p^r divides both coefficients, and write $d = p^r a$, $e = p^r b$. The pair a, b is not both divisible by p , and

$$f(dn + e) = f(an + b) \quad (n \in I).$$

When $an + b$ has p -adic order zero or one, this is exactly the respective value prescribed by the allowed-word rule. When $p^2 \mid an + b$, including $an + b = 0$, the rule leaves the value unrestricted. Thus a, b witness that the word is allowed. If $(d, e) = (0, 0)$, the word is constantly 1 and has witness $a = 0, b = 1$. Finally $f(1) = 1$ and $f(2) = 2$, so every column is nonconstant. □

For the encoding that follows, parametrize the lines by $x = (x_1, x_2) \in \mathbb{Z}^2$ and write

$$L_n(x) = x_2 + nx_1, \quad u_n = (1, -n) \quad (n \in I). \quad (2.1)$$

The map $L_n: \mathbb{Z}^2 \rightarrow \mathbb{Z}$ is surjective and its kernel is generated by u_n : the equation $x_2 + nx_1 = 0$ is equivalent to $x = x_1(1, -n)$. The model supplies the allowed word $(f(L_n(x)))_{n \in I}$ at every x . Moreover, if $x_1 \equiv 0$ and $x_2 \equiv t \pmod{p}$ for $t \in \{1, 2\}$, every entry of this word equals t . Indeed, every $L_n(x)$ then has the nonzero residue t , so its last nonzero digit is t .

3 Encoding symbols in a cyclic finite coordinate

We now express the word constraints by finitely many translational tiling equations on a common unknown set. A symbol will be represented by the dependence of a function on one of its input digits. The present section constructs tests forbidding specified simultaneous dependences; the next section will force enough dependence to obtain an array. Encoding functional constraints by tiling equations is developed in Greenfeld and Tao [4, Sections 4–8] and Greenfeld and Tao [5, Theorem 4.1]. The issue here is to carry out the required tests with a cyclic finite factor; the graph, dependence, and cycle constructions below supply their exact local forms.

3.1 Channels and the ambient group

Retain the prime $p > 200$, the width $N = p^2$, and the alphabet $\Sigma = \mathbb{F}_p^\times$ from the preceding section. Recall the line maps $L_n(x) = x_2 + nx_1$ and their kernel generators $u_n = (1, -n)$ from (2.1). The set of *channels* is

$$\mathcal{I} = \{0, \dots, N-1\} \sqcup \{\eta_1, \eta_2\}.$$

The first N channels are ordinary; the last two are seeds. Their input digit labels are $J_n = \Sigma$ and $J_{\eta_t} = \{*\}$, respectively. The seeds will eventually require the ordinary channels to display the constant words 1 and 2 at some points. Set

$$S = (\mathbb{Z}/p^2\mathbb{Z})^2, \quad s(x) = x \bmod p^2, \quad D = |S| = p^4.$$

Here S records residues of the horizontal variable x .

For each $i \in \mathcal{I}$, choose primes a_i, b_i and put $P_i = \mathbb{Z}/a_i\mathbb{Z} \times \mathbb{Z}/b_i\mathbb{Z}$. Require

$$a_{\eta_1} = 2, \quad a_{\eta_2} = 3, \quad b_{\eta_1}, b_{\eta_2} > D,$$

and choose all the a_i, b_i to be distinct and different from p . For each $j \in J_i$, choose a further prime r_{ij} , with all these primes distinct and disjoint from the previous choices, such that

$$r_{ij} = A_{ij}a_i + B_{ij}b_i, \quad A_{ij}, B_{ij} \in \mathbb{Z}_{\geq 0}. \quad (3.1)$$

These choices are possible using only finitely many primes. Indeed, if a, b are coprime and $r \geq (a-1)b$, choose $B \in \{0, \dots, a-1\}$ with $Bb \equiv r \pmod{a}$; then $A = (r - Bb)/a$ is a nonnegative integer. There are arbitrarily large primes outside any prescribed finite set, so this argument applies successively to every required r_{ij} .

Write

$$r_i = \prod_{j \in J_i} r_{ij}, \quad K_i = \mathbb{Z}/r_i\mathbb{Z} \cong \prod_{j \in J_i} \mathbb{Z}/r_{ij}\mathbb{Z}, \quad K = \prod_{i \in \mathcal{I}} K_i, \quad P = \prod_{i \in \mathcal{I}} P_i.$$

The displayed identification of K_i is the Chinese remainder isomorphism, given by taking residues. Let $U_{ij} \leq K_i$ be the coordinate subgroup supported on the digit j in this identification. We work in

$$G = \mathbb{Z}^2 \times V, \quad V = \mathbb{Z}/D\mathbb{Z} \times \prod_{i \in \mathcal{I}} (\mathbb{Z}/r_i^2\mathbb{Z} \times P_i). \quad (3.2)$$

The group V is cyclic. To see this, expand each P_i into its two cyclic factors. The orders D , r_i^2 , a_i , and b_i of all the resulting factors are pairwise coprime. A tuple of generators therefore has order equal to their product, which is $|V|$. The residue set S is not an additional factor of V ; the coordinate of order D in (3.2) is the cyclic group $\mathbb{Z}/D\mathbb{Z}$.

3.2 One graph and freely chosen high coordinates

Write an element of G as

$$(x, z, (v_i, c_i)_{i \in \mathcal{I}}), \quad z \in \mathbb{Z}/D\mathbb{Z}, \quad v_i \in \mathbb{Z}/r_i^2\mathbb{Z}, \quad c_i \in P_i.$$

Its low coordinates are $k_i = v_i \bmod r_i \in K_i$. Let $q_0 : G \rightarrow \mathbb{Z}^2 \times K$ be the homomorphism retaining just (x, k) , and let

$$H_0 = \ker q_0 = \{0\} \times \mathbb{Z}/D\mathbb{Z} \times \prod_{i \in \mathcal{I}} (r_i(\mathbb{Z}/r_i^2\mathbb{Z}) \times P_i).$$

Include the tiling equation

$$A \oplus H_0 = G. \quad (3.3)$$

For a target point in any fibre of q_0 , every element of A in that fibre supplies exactly one representation with a summand in H_0 . Consequently (3.3) says precisely that A contains one point above each (x, k) . We may thus denote its outputs by $c_i(x, k)$, $z(x, k)$, and $v_i(x, k)$.

For subsequent constructions it is useful to describe v_i by a sheared high coordinate $\beta_i(x, k) \in K_i$, defined by

$$v_i(x, k) = x_1 + [k_i - x_1]_{r_i} + r_i \beta_i(x, k) \quad \text{in } \mathbb{Z}/r_i^2\mathbb{Z}. \quad (3.4)$$

Here $[\cdot]_{r_i}$ is the least nonnegative representative modulo r_i . For fixed x, k_i , the right side is a bijection from K_i onto the elements of $\mathbb{Z}/r_i^2\mathbb{Z}$ with residue k_i . Thus arbitrary functions c_i , β_i , and z specify exactly one graph satisfying (3.3). This is a choice of coordinates on each fibre; it does not assert a direct-product splitting of $\mathbb{Z}/r_i^2\mathbb{Z}$ into two groups of order r_i .

If we add $h = (h_1, h_2)$ to x and h_1 to v_i , the new low coordinate is $k_i + h_1$. Since

$$[(k_i + h_1) - (x_1 + h_1)]_{r_i} = [k_i - x_1]_{r_i},$$

this operation preserves β_i . The identity holds for every integer lift of h_1 , including negative ones.

3.3 Dependence constraints on the useful outputs

We require exactly the following restrictions on the functions c_i :

$$\begin{aligned} c_n(x, k) &= \tilde{c}_n(L_n(x), k_n) \quad (0 \leq n < N), \\ c_{\eta_t}(x, k) &= \tilde{c}_{\eta_t}(s(x), k_{\eta_t}) \quad (t = 1, 2). \end{aligned} \quad (3.5)$$

The outputs c_i will encode symbols; the other outputs will be used to satisfy the activation equations.

Lemma 3.1. *Fix a channel i and a shift $\sigma \in \mathbb{Z}^2 \times K$. There is a finite nonempty tile $F_{i,\sigma} \subset G$ such that, for every graph satisfying (3.3),*

$$A \oplus F_{i,\sigma} = G \iff c_i(b - \sigma) = c_i(b) \text{ for every } b \in \mathbb{Z}^2 \times K.$$

This equation imposes no restriction on the graph's other outputs.

Proof. Choose any lift $g \in G$ of σ with c_i -coordinate zero, and partition H_0 into

$$E_0^{(i)} = \{h \in H_0 : c_i(h) = 0\}, \quad E_{\neq 0}^{(i)} = H_0 \setminus E_0^{(i)}.$$

Set

$$F_{i,\sigma} = E_{\neq 0}^{(i)} \sqcup (g + E_0^{(i)}).$$

The union is disjoint because its two parts have, respectively, nonzero and zero c_i -coordinates. Fix a target base b . The unique graph point above b , together with $E_{\neq 0}^{(i)}$, covers exactly those points in that fibre whose c_i -coordinate differs from $c_i(b)$, once each. The graph point above $b - \sigma$, together with $g + E_0^{(i)}$, covers exactly the points whose c_i -coordinate is $c_i(b - \sigma)$, once each. In both assertions, every choice of the other outputs and high coordinates is attained exactly once, since those offsets range freely in $E_0^{(i)}$. The two contributions partition the target fibre precisely when the displayed equality of useful outputs holds. $\square$

Apply Theorem 3.1, for each i , to a generator of every K_ℓ with $\ell \neq i$, keeping all other base coordinates fixed. This removes dependence on the other low inputs. For an ordinary channel n , also apply it to the horizontal shift u_n with zero low-coordinate shift. Since $\ker L_n = \mathbb{Z}u_n$, the remaining dependence on x factors through $L_n(x)$. For each seed use instead the two horizontal shifts $(p^2, 0)$ and $(0, p^2)$, again with zero low-coordinate shift. These give precisely dependence on $s(x)$. There are only finitely many such generators and tiles, and their equations are equivalent to (3.5). In particular, they leave all the functions β_i and z unrestricted.

For a graph with these dependences, write $g_{i,x} : K_i \rightarrow P_i$ for its useful output at x . Define its *active labels* by

$$\mathcal{A}_i(x) = \{j \in J_i : \text{there exist } w, w' \in K_i \text{ with } w - w' \in U_{ij} \text{ and } g_{i,x}(w) \neq g_{i,x}(w')\}.$$

Thus a label is inactive exactly when changing that digit never changes the output. For ordinary n , the set $\mathcal{A}_n(x)$ depends only on $L_n(x)$; for a seed it depends only on $s(x)$. A function independent of every digit is constant, since one can pass between any two inputs by changing one digit at a time. At this stage the active sets are allowed to be empty.

3.4 A cycle test for simultaneous activity

Lemma 3.2 (Exclusion of simultaneous activity). *Let $C = (i_1, \dots, i_t)$ be a nonempty cyclically ordered list of distinct channels, and choose a label $j(i) \in J_i$ for each $i \in C$. There is a finite family of finite nonempty tiles such that a graph satisfying (3.5) tiles with every member of this family if and only if, at every $x \in \mathbb{Z}^2$, at least one of the chosen labels $j(i)$ is inactive.*

Proof. For each $i \in C$, let $\text{prev}(i)$ denote the preceding channel in the cycle. Choose arbitrary maps

$$d_i : P_{\text{prev}(i)} \longrightarrow U_{i,j(i)}.$$

For $e = (e_i)_{i \in \mathcal{I}} \in P$, put

$$d_i[e] = \begin{cases} d_i(e_{\text{prev}(i)}) & i \in C, \\ 0 & i \notin C. \end{cases}$$

Include a tile for every joint choice $d = (d_i)_{i \in C}$, namely

$$F_d = \left\{ (0, \zeta, (v_i, e_i)_{i \in \mathcal{I}}) : \begin{array}{l} e \in P, \quad \zeta \in \mathbb{Z}/D\mathbb{Z}, \\ v_i \in \mathbb{Z}/r_i^2\mathbb{Z}, \quad v_i \bmod r_i = d_i[e] \quad (i \in \mathcal{I}) \end{array} \right\}. \quad (3.6)$$

This is an ordinary set: distinct e give distinct useful-output offsets, and every choice of ζ and the lifts v_i occurs once. Both the tile and the family of all maps d are finite.

Fix a target base (x, k) . For a fixed e , a representation using (3.6) must start at the graph point over $(x, (k_i - d_i[e])_i)$. The useful outputs of the target are then

$$\Phi_{x,k,d}(e) = (e_i + g_{i,x}(k_i - d_i[e]))_{i \in \mathcal{I}}. \quad (3.7)$$

For any prescribed remaining coordinates of the target, the offsets ζ and v_i are uniquely determined: each low residue is already correct, and all its lifts occur in the tile. Hence $A \oplus F_d = G$ holds exactly when (3.7) is a permutation of P for every (x, k) . This criterion is independent of the graph's high outputs and its z -output.

Suppose some $j(i_0)$ is inactive at x . Then the i_0 -component of (3.7) is $e_{i_0} + g_{i_0,x}(k_{i_0})$, independent of the preceding component of e . From any target useful-output tuple, we can therefore recover e_{i_0} uniquely. The next equation around the cycle then recovers the next component, and continuing recovers all components on C . The starting equation remains satisfied because it never depended on the preceding component. Off the cycle, each equation is just a fixed translation and recovers its component separately. This gives a unique preimage for every target, for all k and all maps d .

Conversely, suppose every chosen label is active at some x . For each $i \in C$ choose $w_i, w'_i \in K_i$ such that

$$w_i - w'_i \in U_{i,j(i)}, \quad \varepsilon_i := g_{i,x}(w_i) - g_{i,x}(w'_i) \neq 0.$$

Set $k_i = w_i$ on the cycle, and choose the other k_i arbitrarily. Extend ε by zero off C to an element of P . For each $i \in C$, choose a map with

$$d_i(0) = 0, \quad d_i(\varepsilon_{\text{prev}(i)}) = w_i - w'_i.$$

The two specified arguments are distinct, so these prescriptions extend to maps on the whole domain. They are among the maps tested in the finite family. At this base, the distinct inputs 0 and ε have the same image under (3.7): on C their outputs are respectively $g_{i,x}(w_i)$ and $\varepsilon_i + g_{i,x}(w'_i)$, and off C they coincide as well. The permutation condition fails. This proves both directions. $\square$

Let $\mathcal{W} \subseteq \Sigma^N$ be the finite set of allowed words from the word rule. We impose the following specific instances of Theorem 3.2:

1. For every $w \in \Sigma^N \setminus \mathcal{W}$, use the cycle $(0, 1, \dots, N-1)$ with label $j(n) = w(n)$.
2. For each $t \in \{1, 2\}$, each ordinary channel n , and each $j \in \Sigma \setminus \{t\}$, use the two-channel cycle (η_t, n) with labels $*$ and j .

All these instances form a finite family. Their precise consequences are

$$\prod_{n=0}^{N-1} \mathcal{A}_n(x) \subseteq \mathcal{W}, \quad * \in \mathcal{A}_{\eta_t}(x) \implies \mathcal{A}_n(x) \subseteq \{t\} \quad (0 \leq n < N). \quad (3.8)$$

The first product is identified with a set of words in the natural order of the channels. These statements also hold when some active sets are empty. The activation tiles constructed next will ensure that every ordinary active set is nonempty at every point and that each seed is active somewhere.

Every tile constructed in this section is a finite nonempty subset of G . As a check on the fibre counts, all have cardinality

$$|H_0| = D \prod_{i \in \mathcal{I}} r_i |P_i| :$$

the dependence tiles partition a copy of H_0 into two shifted pieces, while each cycle tile has $|P|$ choices of e , D choices of ζ , and r_i choices of each lift v_i .

4 Forcing activity and excluding full periodicity

The exclusion equations constrain which labels can be active together. We now add one finite tile for each channel to ensure that every ordinary channel is active at every point, and that each seed is active somewhere. Throughout this section, we retain the groups and coordinates of [Section 3](#). In particular, the useful output of channel i is $c_i \in P_i = \mathbb{Z}/a_i\mathbb{Z} \times \mathbb{Z}/b_i\mathbb{Z}$, its low input is $k_i \in K_i$, and its sheared high coordinate is $\beta_i \in K_i$.

4.1 A shift list with two fibre orderings

The shift list will serve two purposes: its nonuniform multiplicities will force activity, while its uniform counts in each coordinate will permit the explicit common solution in [Section 5](#).

Fix a channel i . For $e \in P_i$, let $\mathbf{1}_{\{e\}}$ denote the indicator of the singleton $\{e\}$, and define the integer-valued functions

$$\omega_i = \mathbf{1}_{\{(0,0)\}} - \mathbf{1}_{\{(1,0)\}} - \mathbf{1}_{\{(0,1)\}} + \mathbf{1}_{\{(1,1)\}}, \quad \mu_i = 1 + \omega_i \quad \text{on } P_i.$$

Here 1 denotes the constant function. The function μ_i has values two at $(0,0)$ and $(1,1)$, zero at $(1,0)$ and $(0,1)$, and one elsewhere. It is therefore a nonnegative, nonconstant multiplicity function. Its total mass is $a_i b_i$, and its marginals are uniform:

$$\sum_{\alpha \in \mathbb{Z}/a_i\mathbb{Z}} \mu_i(\alpha, \xi) = a_i \quad (\xi \in \mathbb{Z}/b_i\mathbb{Z}), \quad \sum_{\xi \in \mathbb{Z}/b_i\mathbb{Z}} \mu_i(\alpha, \xi) = b_i \quad (\alpha \in \mathbb{Z}/a_i\mathbb{Z}).$$

Choose an index set Δ_i of size $a_i b_i$ and a list $e_\delta = (e_\delta^a, e_\delta^b) \in P_i$, $\delta \in \Delta_i$, in which each $e \in P_i$ occurs $\mu_i(e)$ times. The following notation for the list and its orderings is local to channel i .

For each ξ , order the a_i indices with $e_\delta^b = \xi$ by an arbitrary bijection to $\mathbb{Z}/a_i\mathbb{Z}$, and define $\rho^a(\delta)$ by subtracting e_δ^a from that assigned value. Independently, for each α , order the b_i indices with $e_\delta^a = \alpha$ by a bijection to $\mathbb{Z}/b_i\mathbb{Z}$, and subtract e_δ^b to define $\rho^b(\delta)$. Thus both maps in

$$\begin{aligned} \{\delta \in \Delta_i : e_\delta^b = \xi\} &\longrightarrow \mathbb{Z}/a_i\mathbb{Z}, & \delta &\longmapsto e_\delta^a + \rho^a(\delta), \\ \{\delta \in \Delta_i : e_\delta^a = \alpha\} &\longrightarrow \mathbb{Z}/b_i\mathbb{Z}, & \delta &\longmapsto e_\delta^b + \rho^b(\delta) \end{aligned} \tag{4.1}$$

are bijections. Repeated values of e_δ have distinct indices; the orderings act on these indices.

4.2 The activation tiles

For a channel ℓ , write $R_\ell = r_\ell \mathbb{Z} / r_\ell^2 \mathbb{Z}$ for the subgroup of offsets with low coordinate zero. It has r_ℓ elements. Given $h \in \mathbb{Z}^2$, $e \in P_i$, and $E \subseteq \mathbb{Z} / D\mathbb{Z}$, define a batch of offsets by

$$\mathcal{B}_i(h, e; E) = \{(h, \zeta, (w_\ell, d_\ell)_{\ell \in \mathcal{I}}) : \zeta \in E, \quad w_i = h_1 \pmod{r_i^2}, \quad d_i = e, \\ w_\ell \in R_\ell, \quad d_\ell \in P_\ell \text{ for every } \ell \neq i\}.$$

Every combination of the freely varied coordinates occurs once in this set.

For an ordinary channel $i \in \{0, \dots, N-1\}$ and each pair $(l, \delta) \in K_i \times \Delta_i$, choose an integer multiple $h = h_{l, \delta}^{(i)}$ of $u_i = (1, -i)$ such that

$$h_1 \equiv l \pmod{r_i}, \quad h_1 \equiv \rho^a(\delta) \pmod{a_i}, \quad h_1 \equiv \rho^b(\delta) \pmod{b_i}. \quad (4.2)$$

The moduli are pairwise coprime, so the Chinese remainder theorem gives infinitely many choices of h_1 . Choose these horizontal offsets distinctly for the different pairs (l, δ) , and set

$$F_i^{\text{act}} = \bigcup_{(l, \delta) \in K_i \times \Delta_i} \mathcal{B}_i(h_{l, \delta}^{(i)}, e_\delta; \mathbb{Z} / D\mathbb{Z}).$$

The restriction $h \in \mathbb{Z}u_i$ gives $L_i(x - h) = L_i(x)$ for every x .

For a seed $i \in \{\eta_1, \eta_2\}$, let i' denote the other seed. For every $(l, \delta, \tau) \in K_i \times \Delta_i \times S$, choose $h = h_{l, \delta, \tau}^{(i)} \in \mathbb{Z}^2$ satisfying Equation (4.2) and the additional conditions

$$s(h) = \tau, \quad h_1 \equiv 0 \pmod{a_{i'}}. \quad (4.3)$$

There is no slope restriction in this case. Writing $\tau = (\tau_1, \tau_2) \in (\mathbb{Z} / p^2 \mathbb{Z})^2$, the first coordinate is prescribed modulo the pairwise coprime integers $r_i, a_i, b_i, p^2, a_{i'}$, while the second coordinate only needs to satisfy $h_2 \equiv \tau_2 \pmod{p^2}$. Again there are infinitely many choices, so take all horizontal offsets within this seed tile to be distinct. Define

$$F_i^{\text{act}} = \bigcup_{(l, \delta, \tau) \in K_i \times \Delta_i \times S} \mathcal{B}_i(h_{l, \delta, \tau}^{(i)}, e_\delta; \{0\}).$$

Thus an ordinary batch varies the z -offset freely, whereas a seed batch has z -offset zero. The congruence involving $a_{i'}$ will allow both seed tests to hold with one common z -output in the construction of Theorem 5.2.

All unions just defined are disjoint unions of finite sets, because different batches have different horizontal offsets. In particular, the repeated entries of the auxiliary shift list do not introduce weighted tile elements. These tiles are nonempty, and direct counting gives, for every channel i ,

$$|F_i^{\text{act}}| = D \prod_{\ell \in \mathcal{I}} r_\ell |P_\ell| = |H_0|.$$

We add the equations $A \oplus F_i^{\text{act}} = G$ to the graph, dependence, and exclusion equations.

4.3 Exact coverage in a fibre

Lemma 4.1 (Activation fibre criterion). *Suppose $A \oplus H_0 = G$, so that A is a graph over (x, k) . For a target base point (x, k) and a batch with horizontal offset h , write*

$$x' = x - h, \quad k'_i = k_i - l, \quad k'_\ell = k_\ell \quad (\ell \neq i). \quad (4.4)$$

For an ordinary channel i , the equation $A \oplus F_i^{\text{act}} = G$ holds if and only if, at every target base point, the map

$$(l, \delta) \mapsto (c_i(x', k') + e_\delta, \beta_i(x', k')) \quad (4.5)$$

is a bijection from $K_i \times \Delta_i$ to $P_i \times K_i$. For a seed i , the corresponding necessary and sufficient condition is that, at every target base point,

$$(l, \delta, \tau) \mapsto (c_i(x', k') + e_\delta, \beta_i(x', k'), z(x', k')) \quad (4.6)$$

be a bijection from $K_i \times \Delta_i \times S$ to $P_i \times K_i \times \mathbb{Z}/D\mathbb{Z}$.

Proof. Fix a target base point and a batch. Every offset in that batch has the same horizontal and low coordinates, so the graph equation supplies exactly one possible source point of A : the point above the base in (4.4). The i th low-coordinate shift is l because $h_1 \equiv l \pmod{r_i}$. Since $x'_1 = x_1 - h_1$, we have

$$[k'_i - x'_1]_{r_i} = [k_i - x_1]_{r_i}.$$

Consequently the shear formula (3.4) gives

$$v_i(x', k') + h_1 = x_1 + [k_i - x_1]_{r_i} + r_i \beta_i(x', k') \quad \text{in } \mathbb{Z}/r_i^2 \mathbb{Z}.$$

Thus the offset preserves the sheared high coordinate β_i exactly, and the target useful output is $c_i(x', k') + e_\delta$.

For every $\ell \neq i$, adding the freely chosen element of R_ℓ fills the entire high-coordinate fibre with low coordinate k_ℓ once. Independently, adding the freely chosen element of P_ℓ fills that useful-output coordinate once. In an ordinary batch the z -coordinate is also filled once; in a seed batch it remains equal to $z(x', k')$. Hence each batch covers exactly the part of the target fibre specified by its tuple in Equation (4.5) or Equation (4.6), with one representation for every choice of the remaining coordinates. Distinct batches correspond to distinct tile elements, so exact coverage by the whole tile is equivalent to every specified tuple occurring once. This is precisely the asserted bijection condition. $\square$

In particular, exact coverage forces every value of the target c_i to occur r_i times among the ordinary batches, or Dr_i times among the seed batches. These necessary marginal counts already force the activity we need.

4.4 Every required channel is active

Lemma 4.2 (Ordinary activation). *Every ordinary channel has at least one active label at every $x \in \mathbb{Z}^2$ in any common solution of the equations specified so far.*

Proof. Suppose channel i has no active label at some x . A function on the finite product $K_i = \prod_{j \in J_i} \mathbb{Z}/r_{ij} \mathbb{Z}$ that is independent of every digit is constant: any two inputs can be joined by changing one digit at a time. Thus $g_{i,x}$ has a constant value $c_0 \in P_i$. By the dependence condition (3.5) and the equality $L_i(x - h) = L_i(x)$, every source in the activation test at (x, k) has $c_i(x', k') = c_0$. For each δ there are r_i choices of l . The number of batches giving any target value $c \in P_i$ is therefore

$$r_i \mu_i(c - c_0).$$

The fibre criterion requires this number to be r_i for every c . That is impossible because μ_i is not constant. $\square$

Lemma 4.3 (Seed activation). *Each of the two seeds has its label active at some point of $\mathbb{Z}^2$ in any common solution of the equations specified so far.*

Proof. Fix a seed i and suppose it is nowhere active. The dependence condition and the absence of activity give a function $q : S \rightarrow P_i$ such that $c_i(x, k) = q(s(x))$ for every base point. Define its nonnegative integer histogram by

$$Q(\alpha, \xi) = |\{s \in S : q(s) = (\alpha, \xi)\}|, \quad \sum_{(\alpha, \xi) \in P_i} Q(\alpha, \xi) = D.$$

At any target base point and for a fixed δ , the source pairs

$$(s(x'), k'_i) = (s(x) - \tau, k_i - l)$$

run once through $S \times K_i$ as (τ, l) varies. Thus the number of seed batches with useful output $c \in P_i$ is $r_i \sum_{\delta \in \Delta_i} Q(c - e_\delta)$. The necessary marginal count from [Theorem 4.1](#), divided by r_i , yields

$$(Q * \mu_i)(c) = D \quad (c \in P_i),$$

where $(Q * \mu_i)(c) = \sum_{e \in P_i} \mu_i(e) Q(c - e)$ is convolution on the finite group P_i . Convolution with the constant function 1 already equals D , so $Q * \omega_i = 0$. Written out, this says

$$Q(\alpha, \xi) - Q(\alpha - 1, \xi) - Q(\alpha, \xi - 1) + Q(\alpha - 1, \xi - 1) = 0. \quad (4.7)$$

We spell out the integer and positivity consequences of this identity. For each α , the difference $Q(\alpha, \xi) - Q(\alpha - 1, \xi)$ is independent of ξ , because successive differences in the second coordinate vanish. Summing such differences along the cyclic first coordinate shows that, for any α_0 ,

$$Q(\alpha, \xi) - Q(\alpha_0, \xi) = Q(\alpha, 0) - Q(\alpha_0, 0).$$

Choose α_0 to minimize $Q(\alpha, 0)$, and put

$$u(\alpha) = Q(\alpha, 0) - Q(\alpha_0, 0), \quad v(\xi) = Q(\alpha_0, \xi).$$

Both functions take nonnegative integer values, and $Q(\alpha, \xi) = u(\alpha) + v(\xi)$. Summing gives

$$D = b_i \sum_{\alpha \in \mathbb{Z}/a_i \mathbb{Z}} u(\alpha) + a_i \sum_{\xi \in \mathbb{Z}/b_i \mathbb{Z}} v(\xi).$$

Since $b_i > D$ and both sums are nonnegative integers, the first sum must vanish. It follows that a_i divides $D = p^4$. But the two seed values of a_i are 2 and 3, whereas $p > 200$ is prime. This contradiction proves the claim. $\square$

4.5 Extraction of the line array

The graph, dependence, exclusion, and activation equations now form a finite system of tiling equations in G . Its solvability will be proved in the next section. We can already exclude all fully periodic common solutions.

Proposition 4.4. *No common solution of this finite system is invariant under a finite-index subgroup of G .*

Proof. Let A be any common solution. Fix a total ordering of Σ . For each ordinary channel n and integer m , define

$$W(n, m) = \min \mathcal{A}_n((0, m)),$$

where the minimum is taken in that ordering. This is defined by [Theorem 4.2](#). The dependence condition implies that $\mathcal{A}_n(x)$ depends only on $L_n(x)$; since $L_n(0, m) = m$, the selected symbol at any x is exactly $W(n, L_n(x))$.

For any integers d, e , consider $x = (d, e)$. If the word $(W(n, dn + e))_{0 \leq n < N}$ were forbidden, all its selected labels would be active at x . This contradicts the corresponding exclusion equations, by [Theorem 3.2](#). Hence W satisfies the line rule.

By [Theorem 4.3](#), for each $t \in \{1, 2\}$ there is a point $x^{(t)}$ where the label of η_t is active. The seed-pair exclusions imply that $\mathcal{A}_n(x^{(t)})$ contains no symbol other than t . Since this set is nonempty, it equals $\{t\}$. Therefore

$$W(n, L_n(x^{(1)})) = 1, \quad W(n, L_n(x^{(2)})) = 2 \quad \text{for every } 0 \leq n < N.$$

The two arguments in each column must be distinct, and every column of W is nonconstant.

Suppose now that A is invariant under a finite-index subgroup $\Lambda \leq G$. Consider the actual group element

$$e_v = ((0, 1), 0_v) \in G.$$

Its coset has finite order in G/Λ , so some integer $M > 0$ satisfies $Me_v \in \Lambda$. We thus obtain a period with *zero finite-coordinate component*:

$$A + ((0, M), 0_v) = A.$$

Translation by this element takes the unique graph point above (x, k) to the unique graph point above $(x + (0, M), k)$ and leaves all useful outputs unchanged. It follows that every active-label set is unchanged under $x \mapsto x + (0, M)$. At $x = (0, m)$ this gives $W(n, m + M) = W(n, m)$ for all n, m . The array is therefore vertically periodic, satisfies the line rule, and has every column nonconstant, contradicting [Theorem 2.4](#). $\square$

5 A common solution of the tiling equations

We now construct one graph satisfying all the equations of [Sections 3](#) and [4](#). The useful output of each ordinary channel will activate exactly the symbol $f(L_i(x))$ from [Theorem 2.5](#). The high coordinates will record positions within labelled blocks. A further labelling of the horizontal residue set S will let the two seed tests use a single common output z .

5.1 Labelled blocks and their inverse map

For every channel i and digit $j \in J_i$, fix a partition of $\mathbb{Z}/r_{ij}\mathbb{Z}$ into subsets of sizes a_i and b_i , using the nonnegative representation of r_{ij} chosen in [Section 3](#). Label each subset of size $d \in \{a_i, b_i\}$ bijectively by $\mathbb{Z}/d\mathbb{Z}$. Copy this same partition and these same labels for every setting of the other digits in $K_i = \prod_{j' \in J_i} \mathbb{Z}/r_{ij'}\mathbb{Z}$. We obtain a partition $\mathcal{B}_{i,j}$ of K_i into labelled blocks. A block fixes all digits except j ; its label records a position in its j -digit subset. The blocks are subsets, not subgroups, and their labels need not respect addition in K_i .

Define a map $C_{i,j} : K_i \rightarrow P_i$ and, for each integer t , a permutation $T_{i,j}(t)$ of K_i as follows. If k_i has label y in a block of size d , put

$$C_{i,j}(k_i) = \begin{cases} (y, 0), & d = a_i, \\ (0, y), & d = b_i. \end{cases}$$

The permutation $T_{i,j}(t)$ sends this point to the point with label $y + t$ in the same block, with addition in $\mathbb{Z}/d\mathbb{Z}$. The map $C_{i,j}$ is independent of every digit except j , since the partitions and labels were copied unchanged over the other digits. It is not independent of j : each block has at least two positions with different useful outputs.

The following inverse calculation is the reason for the fibre orderings in [Equation \(4.1\)](#).

Lemma 5.1 (Block inverse). *Fix i, j , a target horizontal coordinate $x_1 \in \mathbb{Z}$, and a target low coordinate $k_i \in K_i$. For each $(l, \delta) \in K_i \times \Delta_i$, choose an integer $h_1(l, \delta)$ satisfying*

$$h_1(l, \delta) \equiv l \pmod{r_i}, \quad h_1(l, \delta) \equiv \rho^a(\delta) \pmod{a_i}, \quad h_1(l, \delta) \equiv \rho^b(\delta) \pmod{b_i}.$$

Then the map

$$(l, \delta) \mapsto (C_{i,j}(k_i - l) + e_\delta, T_{i,j}(x_1 - h_1(l, \delta))(k_i - l)) \quad (5.1)$$

is a bijection from $K_i \times \Delta_i$ to $P_i \times K_i$.

Proof. Fix a target $((\alpha^*, \xi^*), \beta_i^*)$. The point β_i^* specifies one block $B \in \mathcal{B}_{i,j}$ and its label q . As each $T_{i,j}(t)$ preserves its block, any preimage must have its source $k'_i = k_i - l$ in B .

Suppose first that B has size a_i . Choose δ by the two conditions

$$\begin{aligned} e_\delta^b &= \xi^* && \text{in } \mathbb{Z}/b_i\mathbb{Z}, \\ e_\delta^a + \rho^a(\delta) &= \alpha^* + x_1 - q && \text{in } \mathbb{Z}/a_i\mathbb{Z}. \end{aligned} \quad (5.2)$$

The first fibre ordering makes this choice unique. Set $y = \alpha^* - e_\delta^a$, let k'_i be the unique point of B with label y , and set $l = k_i - k'_i$. The useful output in (5.1) is then (α^*, ξ^*) . The high output has label

$$y + x_1 - h_1(l, \delta) = \alpha^* + x_1 - (e_\delta^a + \rho^a(\delta)) = q \quad \text{in } \mathbb{Z}/a_i\mathbb{Z},$$

so it is β_i^* . Conversely, both target coordinates force exactly these choices of δ, y, k'_i and l .

If B has size b_i , use the second fibre ordering to choose the unique δ satisfying

$$\begin{aligned} e_\delta^a &= \alpha^* && \text{in } \mathbb{Z}/a_i\mathbb{Z}, \\ e_\delta^b + \rho^b(\delta) &= \xi^* + x_1 - q && \text{in } \mathbb{Z}/b_i\mathbb{Z}. \end{aligned} \quad (5.3)$$

Now $y = \xi^* - e_\delta^b$ specifies k'_i in B , and again $l = k_i - k'_i$. The high label is $\xi^* + x_1 - (e_\delta^b + \rho^b(\delta)) = q$ in $\mathbb{Z}/b_i\mathbb{Z}$. These choices are necessary and sufficient, proving the bijection in both cases. $\square$

For an ordinary channel i , put $j_i(x) = f(L_i(x))$ and define

$$c_i(x, k) = C_{i,j_i(x)}(k_i), \quad \beta_i(x, k) = T_{i,j_i(x)}(x_1)k_i. \quad (5.4)$$

Thus its active set is exactly $\{j_i(x)\}$, and c_i has the required dependence on $(L_i(x), k_i)$.

Fix a target base point (x, k) in the activation test for this channel. Every batch has $h \in \mathbb{Z}u_i$, so its source $x' = x - h$ satisfies $L_i(x') = L_i(x)$. Consequently all these sources use the same block partition. Their low coordinates are $k'_i = k_i - l$ and their high outputs are $T_{i,j_i(x)}(x_1 - h_1)k'_i$. The shear (3.4) preserves this high output when the batch adds h_1 to v_i . Hence the batch values (c_i, β_i) are precisely (5.1). By [Theorems 4.1](#) and [5.1](#), the ordinary activation equation is satisfied. Its freely varied z -coordinate places no condition on the common output to be defined next.

5.2 Seed regions and one common output

Choose disjoint subsets $S_{\eta_1}, S_{\eta_2} \subset S$ such that

$$\begin{aligned} |S_{\eta_1}| &= 3, & |S_{\eta_2}| &= 4, \\ S_{\eta_t} &\subset \{(s_1, s_2) \in S : s_1 \equiv 0, s_2 \equiv t \pmod{p}\} \quad (t = 1, 2). \end{aligned} \quad (5.5)$$

Each of the two specified residue classes has p^2 points, so these choices are possible. Write $S_0 = S \setminus (S_{\eta_1} \cup S_{\eta_2})$. Since $p > 200$ is prime, $D = p^4 \equiv 1 \pmod{6}$, and therefore $|S_0| = D - 7$ is divisible by 6. Recall that $a_{\eta_1} = 2$ and $a_{\eta_2} = 3$.

Partition each of these three regions into label spaces according to the following table. A label space is simply a subset equipped with the displayed bijection; these are not assertions about subgroups of S .

Region	Inactive seeds	Label space	Number of copies
S_{η_1}	η_2	$\mathbb{Z}/3\mathbb{Z}$	1
S_{η_2}	η_1	$\mathbb{Z}/2\mathbb{Z}$	2
S_0	η_1, η_2	$\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$	$(D - 7)/6$

For a seed i inactive at s , denote its label coordinate by $y_i(s) \in \mathbb{Z}/a_i\mathbb{Z}$. Thus a point of S is specified by its region, the particular copy of the label space in that region, and its inactive-seed labels.

For $t \in \mathbb{Z}$, define a permutation $R(t)$ of S by preserving each label space and adding t to each of its label coordinates. In particular, $R(t)$ preserves all three regions, and its inverse is $R(-t)$. Fix an arbitrary bijection $\lambda : S \rightarrow \mathbb{Z}/D\mathbb{Z}$ and set

$$z(x, k) = \lambda(R(x_1)s(x)). \quad (5.6)$$

Neither λ nor $R(t)$ is required to be a homomorphism. The set S only indexes the values of this one cyclic output coordinate.

For a seed i , its digit set is the singleton $\{*\}$. Abbreviate $C_{i,*}$ and $T_{i,*}$ to C_i and T_i . Define

$$(c_i(x, k), \beta_i(x, k)) = \begin{cases} (C_i(k_i), T_i(x_1)k_i), & s(x) \in S_i, \\ ((y_i(s(x)), 0), k_i), & s(x) \notin S_i. \end{cases} \quad (5.7)$$

The useful output depends only on $(s(x), k_i)$, as required by (3.5). Its sole digit is active exactly when $s(x) \in S_i$.

5.3 The inverse for each seed test

We verify both seed equations using the same functions (5.6) and (5.7). Fix a seed i , write i' for the other seed, and fix a target base (x, k) . The batch indexed by (l, δ, τ) has a predetermined offset $h = h(l, \delta, \tau)$. Its source satisfies

$$x' = x - h, \quad s' = s(x') = s(x) - \tau, \quad k'_i = k_i - l.$$

In addition to the congruences used by Theorem 5.1, the seed offsets satisfy

$$h_1 \equiv 0 \pmod{a_{i'}}. \quad (5.8)$$

We will recover a unique batch from every target

$$(c_i^*, \beta_i^*, z^*) \in P_i \times K_i \times \mathbb{Z}/D\mathbb{Z}, \quad c_i^* = (\alpha^*, \xi^*).$$

By Theorem 4.1, this is exactly the required tiling property. Put $s'' = \lambda^{-1}(z^*)$. Because $R(x'_1)$ preserves each label space, the points s' and s'' must be in the same region and the same copy of its label space.

The seed is active at the source. Suppose $s'' \in S_i$. In this region the only label coordinate belongs to the other seed i' . Equation (5.8) gives

$$y_{i'}(s'') = y_{i'}(s') + x_1 - h_1 = y_{i'}(s') + x_1 \quad \text{in } \mathbb{Z}/a_{i'}\mathbb{Z}.$$

Thus s' is uniquely determined by s'' : preserve its label-space identity and subtract x_1 from its label. Equivalently, $s' = R(-x_1)s''$ within this region. We now know $\tau = s(x) - s'$. For this fixed τ , apply Theorem 5.1 to the offsets $h_1(l, \delta, \tau)$ and the fixed seed partition. The appropriate formula (5.2) or (5.3) recovers δ ; the source label then fixes k'_i and l . The resulting source has the prescribed s' and sends its z -coordinate to z^* . Every target in this case therefore has exactly one preimage.

The seed is inactive at the source. Suppose $s'' \notin S_i$, and let $q = y_i(s'')$. The source high coordinate equals its low coordinate, so necessarily

$$k'_i = \beta_i^*, \quad l = k_i - \beta_i^*.$$

The target useful output and the rotated i -label impose

$$e_\delta^b = \xi^*, \quad y_i(s') = \alpha^* - e_\delta^a, \quad q = y_i(s') + x_1 - \rho^a(\delta).$$

Consequently δ is the unique index satisfying (5.2). These equations then fix the source label $y_i(s')$. If the other seed is also inactive in this region, its source label is uniquely fixed by

$$y_{i'}(s') = y_{i'}(s'') - x_1 \quad \text{in } \mathbb{Z}/a_{i'}\mathbb{Z},$$

using (5.8). If the other seed is active, there is no second label to recover. Keeping the already known label-space identity, these labels determine exactly one s' , and then $\tau = s(x) - s'$ fixes the remaining batch index. The offset for this recovered batch satisfies all the congruences used in the calculation, so it produces exactly the desired target. The recovery uses only the prescribed residues of h_1 , so the offset indexed by the recovered (l, δ, τ) has exactly the label action used to determine those indices. There is no dependence on a later choice of an integer lift. The necessity of every recovery step proves uniqueness.

For emphasis, on S_0 both labels are recovered on the same label space. In the test of i their forward values are

$$\begin{aligned} y_i(s'') &= \alpha^* + x_1 - (e_\delta^a + \rho^a(\delta)), \\ y_{i'}(s'') &= y_{i'}(s') + x_1. \end{aligned}$$

The first line selects δ by the fibre ordering and then fixes $y_i(s')$; the second fixes $y_{i'}(s')$ directly. Interchanging i and i' proves the other seed equation with exactly the same shared output z . Each equation has its own unique representing batch for a target; there is no requirement that those two batches coincide.

5.4 Compatibility of all the equations

The definitions (5.4), (5.6), and (5.7) specify $c_i(x, k)$ and $\beta_i(x, k)$ for every channel, as well as $z(x, k)$, at every base point (x, k) . By (3.4), each chosen β_i determines exactly one $v_i \in \mathbb{Z}/r_i^2\mathbb{Z}$ with low coordinate k_i . Taking these points for all (x, k) defines a single set $A \subset G$ satisfying the graph equation $A \oplus H_0 = G$.

The inverse calculations hold for every integer x_1 , including negative values, since all rotations use addition in the specified finite label groups. They use only the stated residues of h_1 , together

with $s(h) = \tau$ in a seed test and $L_i(h) = 0$ in an ordinary test. Thus any choices of the permitted Chinese remainder lifts, including those made to separate distinct batches, give the same verification. The actual integer x_1 in the high outputs and in z causes no extra constraint: the dependence equations (3.5) restrict only the useful outputs c_i , not β_i or z . The useful outputs have exactly those prescribed dependencies, and all activation equations have just been checked on this one graph.

It remains to check the exclusion equations. At every x , the unique active ordinary word is

$$(f(L_0(x)), \dots, f(L_{N-1}(x))),$$

which is allowed by Theorem 2.5. Hence no forbidden ordinary word has all its labels active. If the seed η_t is active at x , then (5.5) gives $L_n(x) \equiv t \pmod{p}$ for every ordinary n . Since $t \in \{1, 2\}$ is nonzero modulo p , we have $f(L_n(x)) = t$. No forbidden seed and ordinary-symbol pair is simultaneously active either. Theorem 3.2 now verifies every exclusion tile.

Combining this common solution with the obstruction already proved in Theorem 4.4 yields the required intermediate result.

Theorem 5.2. *There are a finite cyclic group V and finitely many nonempty finite sets $F_1, \dots, F_s \subset G = \mathbb{Z}^2 \times V$ for which the simultaneous tiling equations*

$$A \oplus F_\nu = G \quad (1 \leq \nu \leq s)$$

have a common solution, but no common solution is invariant under a finite-index subgroup of G .

Proof. Use the graph, dependence, exclusion, and activation tiles constructed in Sections 3 and 4. They are finite nonempty sets, and their finite coordinate group V is cyclic. The graph constructed in this section solves all their equations. Theorem 4.4 excludes every fully periodic common solution. $\square$

6 Stacking the system into one tile

We now replace the finite system of Theorem 5.2 by one translational tiling equation. The extra finite coordinate must preserve cyclicity, so we use a fresh prime cyclic group. The argument is a version of the stacking construction in Greenfeld and Tao [5, Theorem 3.1 and Lemma 3.2], with earlier consolidation results in Greenfeld and Tao [4, Theorem 1.15 and Section 3]. We give the partition and uniqueness arguments in full, choosing a fresh prime to retain the cyclic finite factor.

Lemma 6.1 (A partition with full difference sets). *For every positive integer s and every finite set of primes, there is a prime q outside that set and a partition*

$$\mathbb{Z}/q\mathbb{Z} = E_1 \sqcup \dots \sqcup E_s$$

into nonempty sets satisfying $E_\nu - E_\nu = \mathbb{Z}/q\mathbb{Z}$ for every ν .

Proof. Choose a sufficiently large odd prime q , and color the elements of $\mathbb{Z}/q\mathbb{Z}$ independently and uniformly with s colors. Fix a nonzero $d \in \mathbb{Z}/q\mathbb{Z}$ and a color ν . Since q is prime, the elements $0, d, 2d, \dots, (q-1)d$ form one cycle. The pairs

$$(0, d), (2d, 3d), \dots, ((q-3)d, (q-2)d)$$

are disjoint, and the second element minus the first is d in every pair. The probability that a given pair has color ν at both ends is s^{-2} ; these events are independent across the pairs. Therefore the probability that $d \notin E_\nu - E_\nu$ is at most

$$(1 - s^{-2})^{\lfloor q/2 \rfloor}.$$

By the union bound, the probability that some color fails to realize some nonzero difference is at most

$$s(q-1)(1-s^{-2})^{\lfloor q/2 \rfloor}.$$

For $s = 1$ this is zero, and for fixed $s > 1$ it tends to zero as q tends to infinity. Choose q large enough that the bound is less than one, also avoiding the specified finite set of primes. A successful coloring gives every nonzero difference in every color. Each color class is consequently nonempty, which also gives the zero difference. $\square$

Proposition 6.2 (Stacking and periodicity). *Let H be an abelian group, let $F_1, \dots, F_s \subset H$ be finite nonempty sets, and let $E_1, \dots, E_s$ be a partition as in [Theorem 6.1](#). Define*

$$F^{\text{st}} = \bigcup_{\nu=1}^s (F_\nu \times E_\nu) \subset H \times \mathbb{Z}/q\mathbb{Z}.$$

There is a common solution to $A \oplus F_\nu = H$ for all ν if and only if F^{st} tiles $H \times \mathbb{Z}/q\mathbb{Z}$. Moreover, a fully periodic tiling by F^{st} gives a fully periodic common solution, and a fully periodic common solution gives a fully periodic tiling by F^{st} .

Proof. If A is a common solution, then $A \times \{0\}$ is a tiling set for F^{st} : for a target (g, t) , the partition of $\mathbb{Z}/q\mathbb{Z}$ selects a unique ν with $t \in E_\nu$, and the equation $A \oplus F_\nu = H$ selects a unique $g = a + f$. If A has a finite-index period subgroup $P \leq H$, then $P \times \{0\}$ has finite index in $H \times \mathbb{Z}/q\mathbb{Z}$ and preserves this tiling set.

Conversely, suppose that $B \oplus F^{\text{st}} = H \times \mathbb{Z}/q\mathbb{Z}$, and fix $g \in H$. A contribution to the fibre $\{g\} \times \mathbb{Z}/q\mathbb{Z}$ of color ν consists of a pair $((a, t), f)$ with $(a, t) \in B$, $f \in F_\nu$, and $a + f = g$. This contribution covers the vertical set $t + E_\nu$. There are only finitely many contributions: a is determined by f , and there are at most q possible values of t . Two distinct contributions of the same color cannot occur. Indeed, the identity $E_\nu - E_\nu = \mathbb{Z}/q\mathbb{Z}$ makes any two translates of E_ν intersect, contradicting uniqueness in the stacked tiling.

Let $n_\nu(g) \in \{0, 1\}$ be the number of contributions of color ν . Exact coverage of the fibre gives

$$\sum_{\nu=1}^s n_\nu(g) |E_\nu| = q = \sum_{\nu=1}^s |E_\nu|. \quad (6.1)$$

Every $|E_\nu|$ is positive, so [Equation \(6.1\)](#) forces $n_\nu(g) = 1$ for all ν .

Projection $\text{pr} : H \times \mathbb{Z}/q\mathbb{Z} \rightarrow H$ is injective on B . Otherwise, two distinct points $(a, t), (a, t') \in B$, together with any $f \in F_1$, would give two color-1 contributions at $g = a + f$. Consequently $A = \text{pr}(B)$ is a set of projected translations without multiplicity. The equality $n_\nu(g) = 1$ now says exactly that g has one representation in $A + F_\nu$. Thus $A \oplus F_\nu = H$ for every ν .

Finally, if $B + P = B$ for a finite-index subgroup $P \leq H \times \mathbb{Z}/q\mathbb{Z}$, then $A + \text{pr}(P) = A$. The induced map

$$(H \times \mathbb{Z}/q\mathbb{Z})/P \longrightarrow H/\text{pr}(P)$$

is surjective, so $\text{pr}(P)$ has finite index. This proves the periodicity assertion. $\square$

Corollary 6.3 (One tile with a cyclic finite coordinate). *For some positive integer Q , there is a finite nonempty set $F \subset \Gamma = \mathbb{Z}^2 \times \mathbb{Z}/Q\mathbb{Z}$ which tiles Γ but admits no fully periodic tiling.*

Proof. Apply [Theorem 6.2](#) to the finite system from [Theorem 5.2](#) in $G = \mathbb{Z}^2 \times V$. Choose the prime q in [Theorem 6.1](#) not to divide $|V|$. Since V is cyclic, $V \times \mathbb{Z}/q\mathbb{Z}$ is cyclic of order $Q = q|V|$: a pair of generators has order Q by coprimality. The common solution gives a tiling by the stacked

tile, and a fully periodic stacked tiling would give a fully periodic common solution, contrary to [Theorem 5.2](#). Identify the finite product with $\mathbb{Z}/Q\mathbb{Z}$ and call the stacked tile F . Since every system tile has cardinality $M = |H_0|$, the disjoint stacking gives $|F| = qM$. $\square$

7 Passing to three-dimensional space

It remains to remove the finite coordinate in [Theorem 6.3](#). We construct a finite tile in $\mathbb{Z}^3$ whose unit-cube thickening also excludes periodic tilings with arbitrary real translation vectors. Quotient-to-lattice reductions are studied by Meyerovitch et al. [8, Theorem 1.1], building on earlier rigid tile constructions such as Greenfeld and Tao [4, Lemma 9.3]. The need to constrain real translation vectors also appears in the Euclidean reduction of Greenfeld and Tao [5, Theorem 2.1 and Lemma 2.2]. The argument below proves both required statements directly for the same finite tile and its unit-cube thickening.

Let

$$\pi : \mathbb{Z}^3 \longrightarrow \Gamma, \quad \pi(a_1, a_2, a_3) = (a_1, a_2, a_3 \bmod Q), \quad w = (0, 0, Q).$$

Thus $\ker \pi = \mathbb{Z}w$. Translate the tile F of [Theorem 6.3](#) so that $0 \in F$, and choose a finite set $U \subset \mathbb{Z}^3$ containing 0 on which π is a bijection onto F . In particular,

$$U \cap \ker \pi = \{0\}, \quad w \notin U. \quad (7.1)$$

Translating F changes neither tileability nor the existence of a fully periodic tiling.

7.1 A residue transversal with many differences

Lemma 7.1 (Rigid representatives). *For a sufficiently large integer $m \geq 2$, one can choose*

$$T_0 = \{t(v) : v \in \{0, \dots, m-1\}^3\}, \quad t(v) \equiv v \pmod{m},$$

such that, with $T_ = T_0 \setminus \{t(0)\}$,*

$$T_* - T_* \supset \{d \in \mathbb{Z}^3 : \|d\|_\infty \leq m, d \notin m\mathbb{Z}^3\}. \quad (7.2)$$

Proof. Choose independently, for each residue v ,

$$t(v) = v + mh_v, \quad h_v \text{ uniform in } \{-2, -1, 0, 1, 2\}^3.$$

Fix a requested vector d on the right-hand side of [Equation \(7.2\)](#). Translation by $d \bmod m$ on $(\mathbb{Z}/m\mathbb{Z})^3$ is a permutation without fixed points. A cycle of length $L \geq 2$ contains $\lfloor L/2 \rfloor \geq L/3$ disjoint successive pairs. Selecting these pairs in every cycle gives at least $m^3/3$ disjoint pairs of residues (u, v) with $v - u \equiv d \pmod{m}$. Discard the pair containing residue 0, if there is one. At least

$$k_m = \lfloor m^3/3 \rfloor - 1$$

pairs remain, all with both endpoints nonzero.

For such a pair, the equality $t(v) - t(u) = d$ asks for

$$h_v - h_u = \frac{d - (v - u)}{m}.$$

This is an integer vector whose coordinates have absolute value at most 2, since $\|d\|_\infty \leq m$ and the coordinates of u, v lie between 0 and $m-1$. Two independent uniform variables on $\{-2, -1, 0, 1, 2\}$

have any specified difference of absolute value at most 2 with probability at least $3/25$. Hence each chosen pair realizes d with probability at least

$$c = (3/25)^3 = 27/15625.$$

These events are independent because the pairs have disjoint endpoints. Failure to realize this d has probability at most $(1 - c)^{k_m}$. There are at most $(2m + 1)^3$ requested vectors, so the probability that Equation (7.2) fails is at most

$$(2m + 1)^3(1 - c)^{k_m}.$$

This tends to zero as m tends to infinity. Fix $m \geq 2$ for which it is less than one, and then fix a successful choice of the representatives. Every witnessing pair avoids residue 0, as required. $\square$

Define the second transversal by moving just the marked point:

$$T_1 = T_* \cup \{t(0) + mw\}.$$

Both T_0 and T_1 have exactly one representative of each residue modulo m . The switch is illustrated schematically in Figure 2.

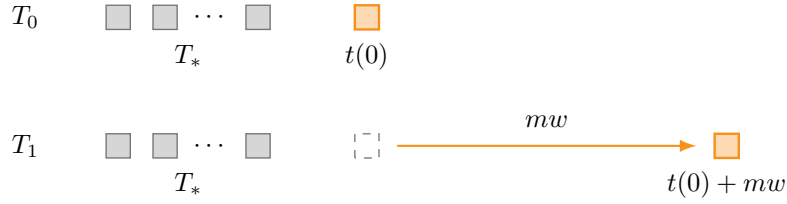


Figure 2: Schematic of the marked-point switch, without depicting the actual positions or scale. The gray marks represent the same common set T_* in both rows. The marked point moves from $t(0)$ to $t(0) + mw$; the dashed outline shows its former location. The marks may also denote unit-cube thickenings. In a grid assembly this switch gives the kernel-invariance identity (7.7).

Our final discrete tile is

$$T = T_1 \cup \bigcup_{u \in U \setminus \{0\}} (mu + T_0), \quad \Omega = T + [0, 1]^3. \quad (7.3)$$

The finite union defining T is disjoint. To see this, equality of two points first forces equality of their residue labels v modulo m . For $v \neq 0$, the points are $mu + t(v)$, and equality forces the same u . For $v = 0$, the unmarked points are $mu + t(0)$ with $u \in U \setminus \{0\}$, while the marked point is $mw + t(0)$. They are all distinct by Equation (7.1). In particular, $|T| = |U|m^3$.

Write

$$\Omega_i = T_i + [0, 1]^3 \quad (i = 0, 1), \quad \Omega_* = T_* + [0, 1]^3.$$

Each Ω_i has volume m^3 , and their common subset Ω_* has positive volume $m^3 - 1$. Unit cubes with distinct integer lower corners have disjoint interiors, so the decomposition of T in Equation (7.3) gives the corresponding decomposition of Ω up to null boundaries. We call the translation vector c the center of the copy $c + T_i$ or $c + \Omega_i$.

7.2 Existence of a tiling

Proposition 7.2. *The tile T tiles $\mathbb{Z}^3$, and Ω tiles $\mathbb{R}^3$ up to null sets.*

Proof. Let $A \oplus F = \Gamma$ be a tiling from [Theorem 6.3](#), and put $B = \pi^{-1}(A)$. Then $B + w = B$. Also $B \oplus U = \mathbb{Z}^3$: for any $g \in \mathbb{Z}^3$, the unique representation of $\pi(g)$ in $A + F$ determines a unique $u \in U$, and then $g - u \in B$ is the unique possible other summand.

Place copies of T at all translations in mB . Their constituent copies of T_0 or T_1 have centers $m(b + u)$, for $b \in B$ and $u \in U$. Since $B \oplus U = \mathbb{Z}^3$, there is exactly one such center at every point of $m\mathbb{Z}^3$. Its type is 1 precisely at the centers mB , because the type-1 piece corresponds to $u = 0$.

The assembly with T_0 at every center in $m\mathbb{Z}^3$ tiles $\mathbb{Z}^3$ exactly: each integer vector has a unique residue label v and a unique expression $ma + t(v)$. Switching to type 1 at the centers mB removes the marked points

$$t(0) + mB$$

and inserts

$$t(0) + m(B + w) = t(0) + mB.$$

The insertion is a bijective relocation, so all multiplicities remain one. This proves $mB \oplus T = \mathbb{Z}^3$. Thickening by unit cubes gives an almost-everywhere tiling by Ω with the same translation set: away from the integer coordinate planes, every point belongs to the interior of one integer unit cube. $\square$

7.3 Rigidity for real translation vectors

We next show why arbitrary real translations cannot produce a periodic tiling. The marked-point modification has two roles. The common part forces the centers of the small pieces onto one grid, and the moved point then forces invariance under the kernel direction w .

Lemma 7.3 (Rounding in the closed box). *Let $m \geq 2$, and suppose that $\delta \in \mathbb{R}^3$ satisfies $\|\delta\|_\infty \leq m$ and $\delta \notin m\mathbb{Z}^3$. There is $d \in \mathbb{Z}^3 \setminus m\mathbb{Z}^3$ such that*

$$\|d\|_\infty \leq m, \quad \|\delta - d\|_\infty < 1.$$

Proof. If δ is integral, use $d = \delta$. Otherwise, keep every integer coordinate unchanged and round each noninteger coordinate either down or up. Both choices lie in $[-m, m]$, since the endpoints are integers, and both are at distance strictly less than one from the original coordinate. In at least one noninteger coordinate choose a rounding which is not a multiple of m . Such a choice exists because two consecutive integers cannot both be multiples of $m \geq 2$. The resulting d is not in $m\mathbb{Z}^3$. Coordinates of δ equal to m or $-m$ are left unchanged, so the strict distance estimate also holds on the boundary of the box. $\square$

Lemma 7.4 (A periodic assembly has grid centers). *Suppose copies $c + \Omega_{\iota(c)}$, with types $\iota(c) \in \{0, 1\}$, form an almost-everywhere tiling of $\mathbb{R}^3$. Assume the assembly, including its types, is invariant under a full-rank lattice $\Lambda \subset \mathbb{R}^3$. Then the set C of centers is $c_0 + m\mathbb{Z}^3$ for some $c_0 \in \mathbb{R}^3$.*

Proof. First, two constituent copies cannot have the same center, because both contain the positive-volume common part Ω_* . Thus the centers and their types are unambiguous. We claim that distinct centers c, c' satisfy

$$\|c' - c\|_\infty \leq m \implies c' - c \in m\mathbb{Z}^3. \quad (7.4)$$

If $\delta = c' - c$ violated this assertion, [Theorem 7.3](#) would give a vector d in the difference set in (7.2) with $\|\delta - d\|_\infty < 1$. Write $d = t - t'$ for $t, t' \in T_*$. The two common-part unit cubes

$$c + t + [0, 1]^3, \quad c' + t' + [0, 1]^3$$

have lower corners whose coordinate differences have absolute value strictly less than one. Their intersection therefore has positive volume, contradicting the tiling property. This proves [Equation \(7.4\)](#).

In particular, distinct centers have sup-norm distance at least m : a nonzero vector in $m\mathbb{Z}^3$ cannot have smaller sup norm. The set C is consequently uniformly separated and locally finite. The auxiliary closed cubes

$$Q_c = c + [-m/2, m/2]^3 \quad (c \in C)$$

have disjoint interiors. Each has volume m^3 , equal to the volume of either constituent type.

We use periodicity to show that these auxiliary cubes cover. Let P be a bounded half-open fundamental parallelepiped for Λ . There are finitely many centers modulo Λ , by local finiteness; choose representatives $c_1, \dots, c_r$. Their types $\iota_1, \dots, \iota_r$ are preserved under Λ . Integrating the constituent tiling over P gives

$$\begin{aligned} |P| &= \int_P \sum_{j=1}^r \sum_{\lambda \in \Lambda} \mathbf{1}_{c_j + \lambda + \Omega_{\iota_j}}(x) dx \\ &= \sum_{j=1}^r |\Omega_{\iota_j}| = rm^3. \end{aligned}$$

For the second equality, translate the integration domains by $-\lambda$ and use that $\{P - \lambda : \lambda \in \Lambda\}$ partitions $\mathbb{R}^3$ up to boundaries. Nonnegative summands justify the exchange of summation and integration.

The same calculation for the auxiliary cubes gives

$$\int_P \sum_{c \in C} \mathbf{1}_{Q_c}(x) dx = rm^3 = |P|.$$

The integrand is at most one almost everywhere, because the cubes have disjoint interiors; their countably many boundaries are null. It is therefore one almost everywhere on P , and periodicity gives almost-everywhere coverage of $\mathbb{R}^3$. The family of closed cubes $\{Q_c\}_{c \in C}$ is locally finite, so its union is closed. A point outside the union would have an open neighborhood of positive volume outside it, contradicting almost-everywhere coverage. Thus the cubes cover every point of $\mathbb{R}^3$.

Their intersection graph is connected. Indeed, if one graph component were separated from the remaining components, the unions of their respective cubes would be disjoint nonempty closed sets covering $\mathbb{R}^3$. Both unions are closed because they are subfamilies of a locally finite family of closed sets. They would disconnect $\mathbb{R}^3$, which is impossible. Whenever Q_c and $Q_{c'}$ intersect, $\|c' - c\|_\infty \leq m$, so [Equation \(7.4\)](#) puts their difference in $m\mathbb{Z}^3$. Following paths in the intersection graph therefore puts every center in one coset $c_0 + m\mathbb{Z}^3$.

No point of that coset can be missing. The interior of the side- m cube at a missing grid point is disjoint from all side- m cubes at the other grid points and would be uncovered. Hence $C = c_0 + m\mathbb{Z}^3$. $\square$

7.4 Kernel invariance and descent

Proposition 7.5. *The Euclidean tile Ω has no almost-everywhere translational tiling whose translation set is invariant under a full-rank lattice in $\mathbb{R}^3$.*

Proof. Suppose, to the contrary, that $\mathcal{A} \subset \mathbb{R}^3$ is such a translation set and that $\mathcal{A} + \Lambda = \mathcal{A}$ for a full-rank lattice Λ . Distinct vectors of $\mathcal{A}$ have sup-norm distance at least one: otherwise

their translates of a fixed unit cube in Ω would overlap in positive volume. Hence $\mathcal{A}$ is countable. Decompose every translated copy of Ω using Equation (7.3). A copy at $a \in \mathcal{A}$ has a type-1 constituent centered at a and type-0 constituents centered at $a + mu$ for $u \in U \setminus \{0\}$. The constituents form an almost-everywhere tiling and inherit the lattice periods, including their types. No two constituents have the same center, as observed in the proof of Theorem 7.4. Consequently the map

$$\mathcal{A} \times U \longrightarrow C, \quad (a, u) \longmapsto a + mu \quad (7.5)$$

is a bijection, and the type-1 centers are exactly $\mathcal{A}$.

By Theorem 7.4, $C = c_0 + m\mathbb{Z}^3$. Translate the entire assembly by $-c_0$, which does not change its period lattice, and henceforth write $C = m\mathbb{Z}^3$. There is a set $B' \subset \mathbb{Z}^3$ such that $\mathcal{A} = mB'$. Let $b(a) = \mathbf{1}_{B'}(a)$; this records the type at the center ma . The bijection (7.5), divided by m , gives

$$B' \oplus U = \mathbb{Z}^3. \quad (7.6)$$

Consider the open unit cell with lower corner $t(0) + ma$, for any $a \in \mathbb{Z}^3$. Since all constituent centers and all voxel positions are integral, a unit cube can meet the interior of this cell only if it has the same lower corner. The residues modulo m then exclude every common-part point $t(v)$ with $v \neq 0$. There are exactly two possible sources: the unshifted marked point from type 0 at ma , or the moved marked point from type 1 at $m(a - w)$. The coverage multiplicity on this open cell is therefore

$$1 - b(a) + b(a - w).$$

The almost-everywhere tiling makes it one. Thus $b(a) = b(a - w)$ for every a , or equivalently

$$B' + w = B'. \quad (7.7)$$

This also gives invariance under every element of $\ker \pi = \mathbb{Z}w$.

The set B' is fully periodic in the discrete sense. It is nonempty, and $\mathcal{A} = mB' \subset m\mathbb{Z}^3$. For $a_0 \in \mathcal{A}$ and any $\lambda \in \Lambda$, both a_0 and $a_0 + \lambda$ belong to $m\mathbb{Z}^3$, so $\lambda \in m\mathbb{Z}^3$. Hence $L = m^{-1}\Lambda$ is a rank-three lattice contained in $\mathbb{Z}^3$ and preserves B' . Such an integer lattice has finite index: the matrix formed by an integer basis has nonzero integer determinant, whose absolute value is its index. In particular, $L \leq \mathbb{Z}^3$ is a finite-index period subgroup of B' .

We finally descend the exact tiling (7.6) to the quotient. Fix $\gamma \in \Gamma$ and any lift $g \in \mathbb{Z}^3$. For each $u \in U$, there is a representation

$$\gamma = \alpha + \pi(u), \quad \alpha \in \pi(B'),$$

if and only if $g - u \in B'$. One direction follows by applying π . For the other, if $\alpha = \pi(b_0)$ with $b_0 \in B'$, then $g - u - b_0 \in \ker \pi$; Equation (7.7) therefore puts $g - u$ in B' . By Equation (7.6), exactly one u has this property, and π is injective on U . Thus

$$\pi(B') \oplus \pi(U) = \Gamma.$$

This quotient tiling is fully periodic. Indeed, $\pi(L)$ preserves $\pi(B')$, and the surjection $\mathbb{Z}^3/L \rightarrow \Gamma/\pi(L)$ shows that $\pi(L)$ has finite index. Since $\pi(U) = F$, this contradicts Theorem 6.3. $\square$

Completion of the proof of Theorem 1.1. The finite nonempty tile T from Equation (7.3) and its closed unit-cube thickening Ω admit the tilings in Theorem 7.2. The Euclidean tiling cannot be fully periodic by Theorem 7.5. If a discrete tiling of T had a finite-index period subgroup in $\mathbb{Z}^3$, thickening its unit cells would give an almost-everywhere tiling of Ω invariant under the same subgroup, viewed as a full-rank lattice in $\mathbb{R}^3$. This is also excluded by Theorem 7.5, proving both claims. $\square$

Remark 7.6 (Finite selection of the tile data). The finite data in the construction can be selected by terminating finite searches. The primes can be selected successively by trial division. The tiling tests involve maps between finite sets. Lemmas 6.1 and 7.1 give explicit inequalities under which a suitable finite coloring or residue transversal exists; exhaustive search then selects one. This describes the finite tile itself. The infinite tiling that proves existence is instead specified by the last-nonzero-digit function. No practical bound on the size of the finite construction is required.

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